

Forecasting Competition, Spring 2026

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0. Introduction

This report aims to forecast a residential customer's daily electricity demand for July 2011 as a class competition. We compared multiple time series models, and evaluated their performance using RMSE and MAPE. Hourly load data from January 2005 to June 2011 are used.

GitHub repository: https://github.com/BellaHuang03/KimHuang_ENV797_TSA_ForecastCompetition_S26

1. Set up

Import datasets

```
head(temp_raw, 10)
```

```
## # A tibble: 10 x 30
##   date                hr t_ws1 t_ws2 t_ws3 t_ws4 t_ws5 t_ws6 t_ws7 t_ws8
##   <dtm>              <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 2005-01-01 00:00:00   1    43    46    40    47    48    46    44    52
## 2 2005-01-01 00:00:00   2    41    46    38    46    48    45    51    50
## 3 2005-01-01 00:00:00   3    40    46    37    45    45    45    49    48
## 4 2005-01-01 00:00:00   4    39    46    37    47    48    48    45    50
## 5 2005-01-01 00:00:00   5    38    46    37    44    48    49    43    50
## 6 2005-01-01 00:00:00   6    37    45    36    45    48    48    40    50
## 7 2005-01-01 00:00:00   7    37    45    37    45    48    44    41    46
## 8 2005-01-01 00:00:00   8    36    43    37    43    48    48    43    46
## 9 2005-01-01 00:00:00   9    36    52    38    49    48    51    51    52
## 10 2005-01-01 00:00:00  10    44    55    43    55    55    55    56    59
## # i 20 more variables: t_ws9 <dbl>, t_ws10 <dbl>, t_ws11 <dbl>, t_ws12 <dbl>,
## #   t_ws13 <dbl>, t_ws14 <dbl>, t_ws15 <dbl>, t_ws16 <dbl>, t_ws17 <dbl>,
## #   t_ws18 <dbl>, t_ws19 <dbl>, t_ws20 <dbl>, t_ws21 <dbl>, t_ws22 <dbl>,
## #   t_ws23 <dbl>, t_ws24 <dbl>, t_ws25 <dbl>, t_ws26 <dbl>, t_ws27 <dbl>,
## #   t_ws28 <dbl>
```

```
head(humid_raw, 10)
```

```
## # A tibble: 10 x 30
```

```
##      date                hr rh_ws1 rh_ws2 rh_ws3 rh_ws4 rh_ws5 rh_ws6 rh_ws7
##      <dtm>                <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
##  1 2005-01-01 00:00:00      1     99    93    93     90     87     93     93
##  2 2005-01-01 00:00:00      2     76    93    97     89     87     97     80
##  3 2005-01-01 00:00:00      3     79    93    93     90     93     97     83
##  4 2005-01-01 00:00:00      4     79    93    93     90     87     89     90
##  5 2005-01-01 00:00:00      5     79    93    96     93     87     90     93
##  6 2005-01-01 00:00:00      6     82    93    97     97     87     93     97
##  7 2005-01-01 00:00:00      7     83    93    90     93     87    100     96
##  8 2005-01-01 00:00:00      8     83    93    88     97     87     93     89
##  9 2005-01-01 00:00:00      9     79    88    93     93     87     89     80
## 10 2005-01-01 00:00:00     10     74    82    89     74     77     80     70
## # i 21 more variables: rh_ws8 <dbl>, rh_ws9 <dbl>, rh_ws10 <dbl>,
## #   rh_ws11 <dbl>, rh_ws12 <dbl>, rh_ws13 <dbl>, rh_ws14 <dbl>, rh_ws15 <dbl>,
## #   rh_ws16 <dbl>, rh_ws17 <dbl>, rh_ws18 <dbl>, rh_ws19 <dbl>, rh_ws20 <dbl>,
## #   rh_ws21 <dbl>, rh_ws22 <dbl>, rh_ws23 <dbl>, rh_ws24 <dbl>, rh_ws25 <dbl>,
## #   rh_ws26 <dbl>, rh_ws27 <dbl>, rh_ws28 <dbl>
```

```
head(load_raw, 10)
```

```
## # A tibble: 10 x 26
##   meter_id date                h1    h2    h3    h4    h5    h6    h7    h8
##   <chr>      <dtm>                <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
##  1 0072      2005-01-01 00:00:00 3072 2911 2784 2768 2832 2904 3064 3192
##  2 0072      2005-01-02 00:00:00 2408 2328 2311 2328 2400 2552 2856 3272
##  3 0072      2005-01-03 00:00:00 2575 2528 2496 2608 2825 3376 3760 3672
##  4 0072      2005-01-04 00:00:00 2177 2016 1951 1913 2080 2624 3328 2896
##  5 0072      2005-01-05 00:00:00 1912 1768 1697 1720 1880 2472 3208 2888
##  6 0072      2005-01-06 00:00:00 2104 1976 1920 1952 2167 2736 3432 3096
##  7 0072      2005-01-07 00:00:00 2336 2296 2336 2448 2712 3408 4248 3943
##  8 0072      2005-01-08 00:00:00 2849 2695 2640 2600 2728 2984 3144 3432
##  9 0072      2005-01-09 00:00:00 3264 3199 3240 3336 3480 3641 4015 4568
## 10 0072      2005-01-10 00:00:00 3520 3568 3584 3728 3943 4680 5256 5000
## # i 16 more variables: h9 <dbl>, h10 <dbl>, h11 <dbl>, h12 <dbl>, h13 <dbl>,
## #   h14 <dbl>, h15 <dbl>, h16 <dbl>, h17 <dbl>, h18 <dbl>, h19 <dbl>,
## #   h20 <dbl>, h21 <dbl>, h22 <dbl>, h23 <dbl>, h24 <dbl>
```

Data Wrangling

The raw temperature, humidity, and load datasets were systematically processed to produce consistent hourly and daily time series for analysis. Timestamps were standardized to POSIXct format, and temperature and humidity readings from multiple weather stations were averaged at the hourly level to obtain representative measures. These hourly series were subsequently aggregated to daily averages. The load dataset was reshaped from wide to long format by transforming hourly columns into a single hour variable, ensuring alignment with the temporal structure of the weather data. The resulting hourly datasets were then merged on common date and hour fields to create a unified dataset. Finally, the merged data were aggregated to the daily level, yielding average daily load, temperature, and humidity variables for subsequent modeling and analysis.

```
knitr::kable(head(daily_data, 10), caption = "First 10 rows of the daily dataset")
```

Table 1: First 10 rows of the daily dataset

date	load_daily	temp_avg	humid_avg
2005-01-01	3107.625	53.56548	76.70982
2005-01-02	3068.292	53.75744	80.47917
2005-01-03	3061.292	55.91369	81.22917
2005-01-04	2565.708	61.69494	74.84226
2005-01-05	2495.708	60.43006	76.06696
2005-01-06	2734.917	62.00446	77.99107
2005-01-07	3179.667	55.85565	71.66667
2005-01-08	3360.708	57.66071	81.80357
2005-01-09	3763.958	47.62351	72.72470
2005-01-10	3559.208	49.29911	81.14435

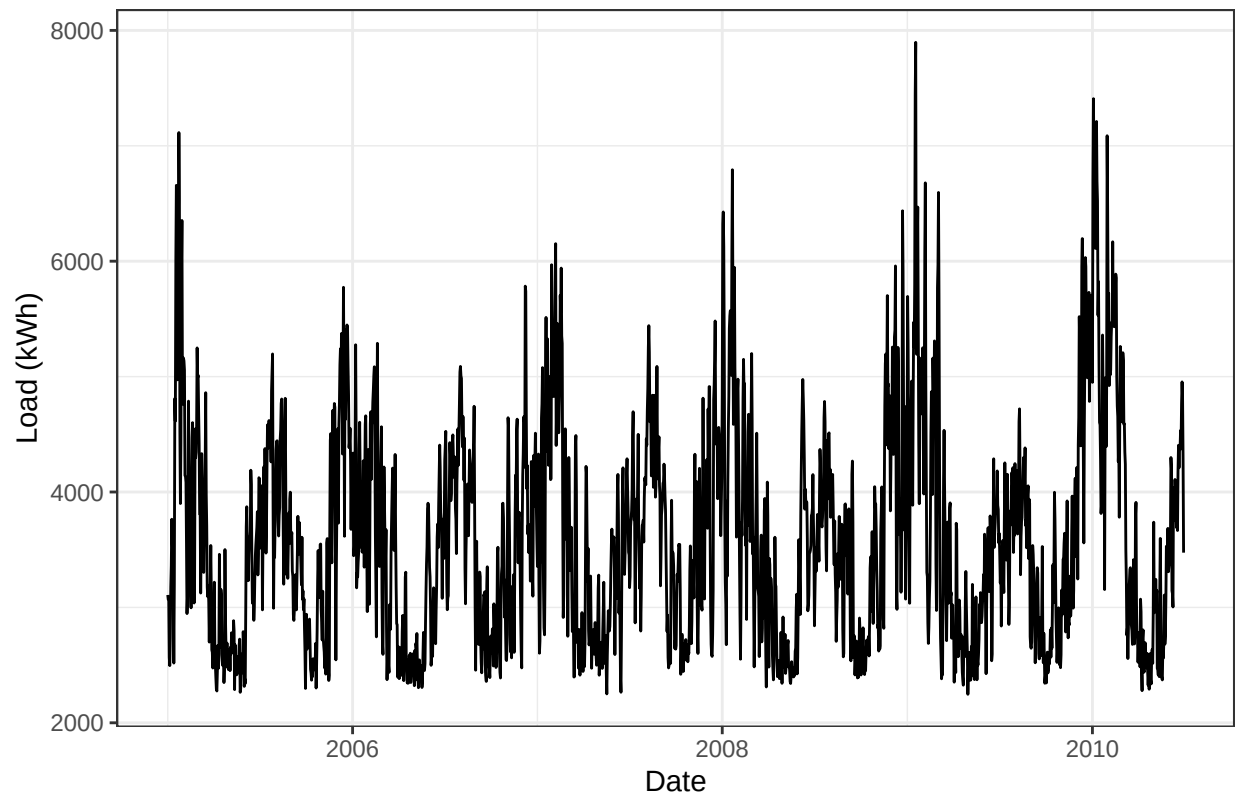
2. Explanatory Data Analysis

An initial time series plot of daily electricity load from January 2005 to June 2010 reveals clear temporal structure characterized by both trend and seasonality. The series exhibits recurring fluctuations, suggesting the presence of strong periodic components, along with an overall upward movement in later years, indicating a gradual increase in load demand over time.

To further decompose these patterns, an STL (Seasonal-Trend decomposition using Loess) was applied with both weekly and annual seasonal periods. The decomposition confirms a pronounced weekly seasonality, likely reflecting differences between weekdays and weekends, as well as a longer annual cycle associated with seasonal demand variations (e.g., weather-driven consumption). The trend component shows a moderate upward trajectory, particularly toward the end of the training period, while the remainder component appears relatively centered around zero, suggesting that most systematic variation has been captured by the trend and seasonal components.

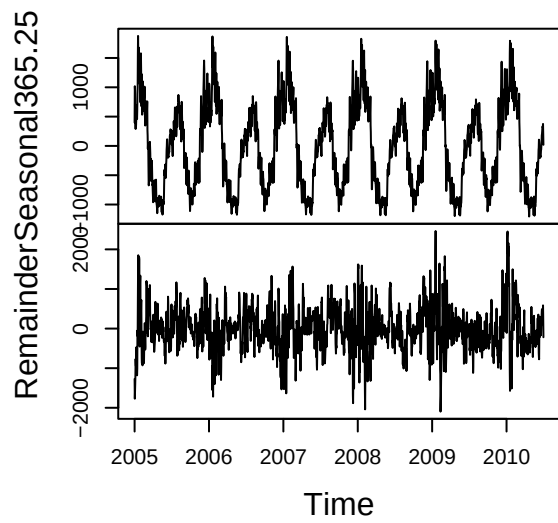
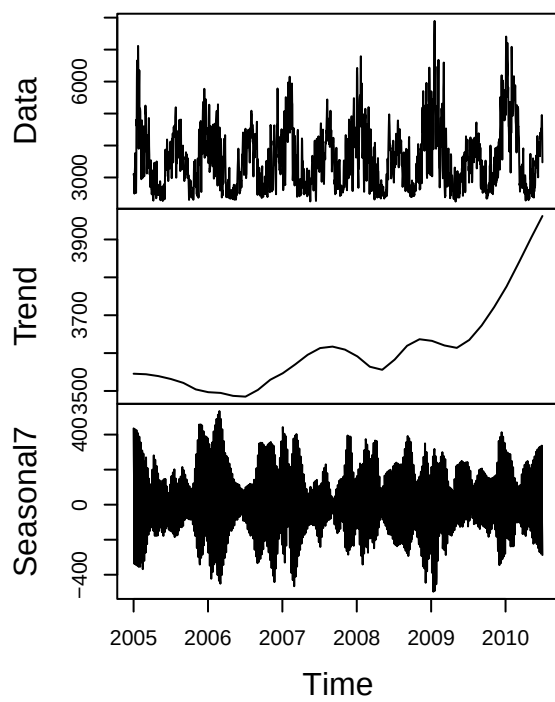
Autocorrelation analysis provides additional insight into the temporal dependence structure. The ACF plot shows a slow decay, indicating strong persistence and non-stationarity in the series, which is consistent with the observed trend and seasonal effects. The PACF plot displays a significant spike at lag 1 followed by smaller contributions at subsequent lags, suggesting that short-term dependencies are important and that lower-order autoregressive components may be sufficient to capture the serial correlation.

Daily Electricity Load – Training Data

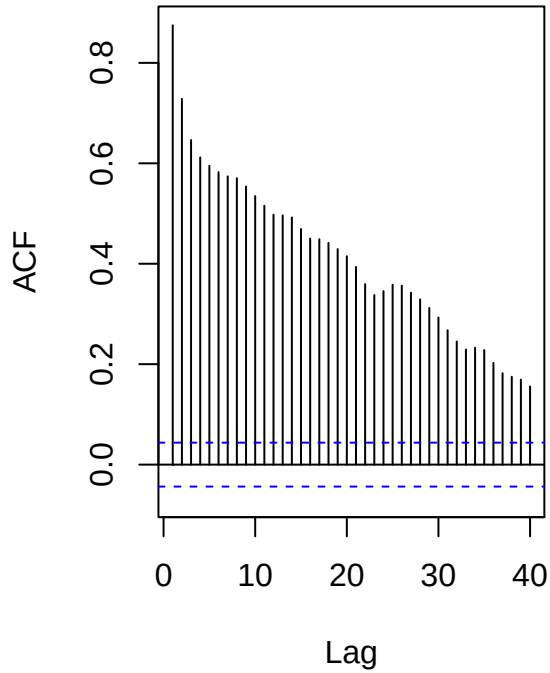


```
##  
## === STL Decomposition ===
```

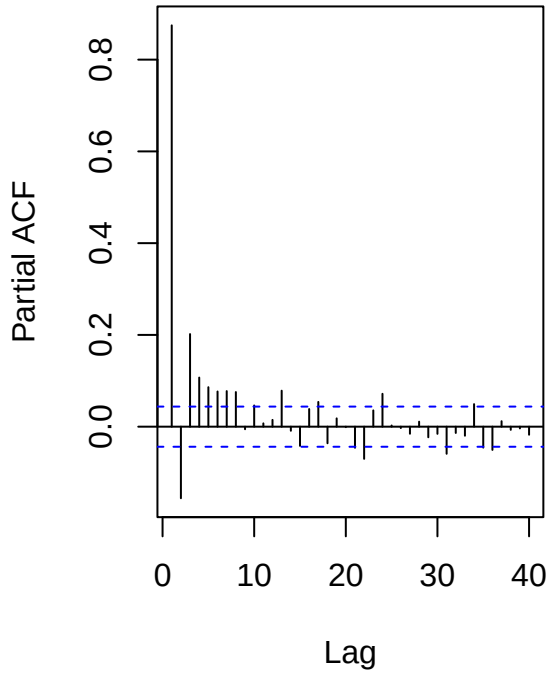
STL Decomposition – Trend + Weekly + Annual + Remainder



Series test_data\$load_daily



Series test_data\$load_daily



3. Top 5 Models

(1) Model 1: TBATS

The TBATS model was selected as an initial forecasting approach due to its ability to handle complex seasonal structures, particularly multiple and non-integer seasonal periods. Given that the electricity load data exhibits both weekly and annual seasonal trends, TBATS provides a flexible framework by incorporating trigonometric representations of seasonality, Box-Cox transformations, and ARMA error structures.

As shown in the results, the TBATS model is able to reasonably track the underlying level and general direction of the series in the validation period. The forecasts reflect the expected seasonal fluctuations and maintain consistency with the recent historical range of the data. However, some deviations between the predicted and actual values are visible, particularly around periods of higher volatility, suggesting that while the model captures broad seasonal patterns effectively, it may be less responsive to short-term irregular spikes.

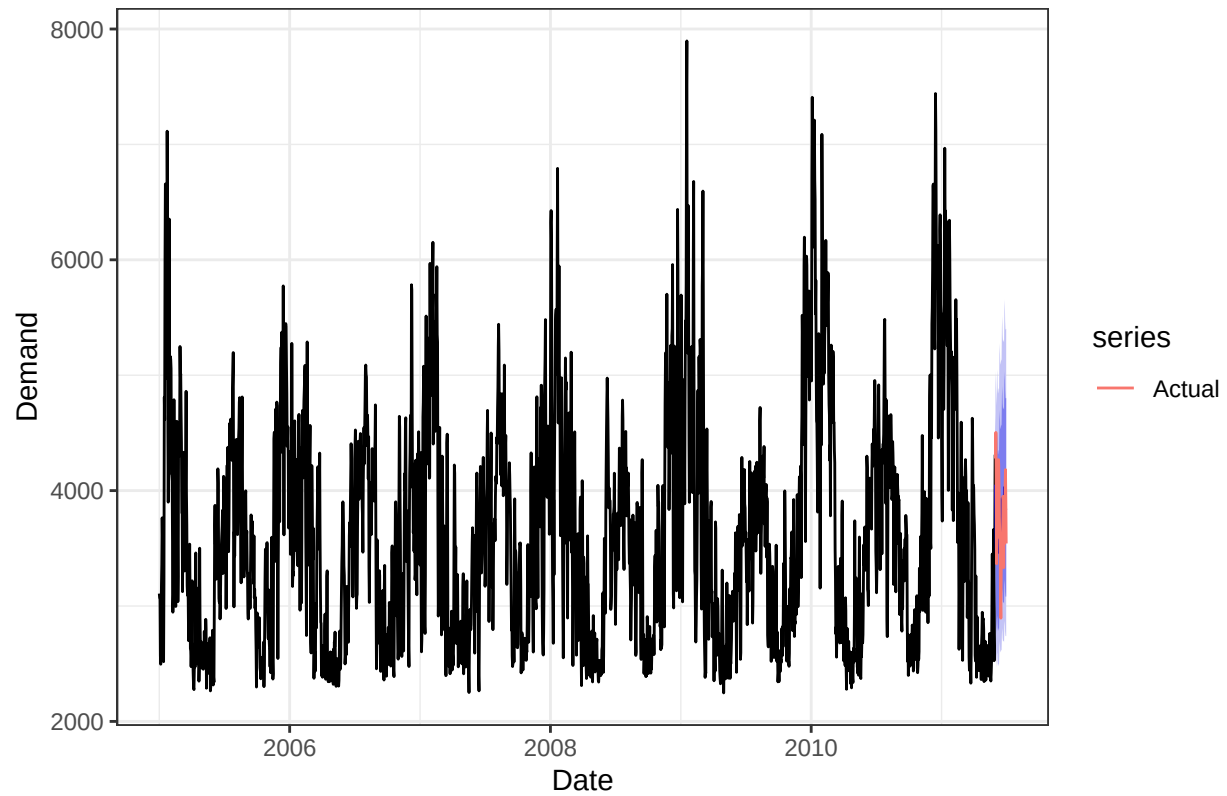
```
## [1] 2341
```

```
## [1] 31
```

```
##           ME      RMSE      MAE      MPE      MAPE      MASE
## Training set 29.72843 428.5227 317.3586 -0.3742386 8.568790 0.4558879
## Test set    45.70463 417.4762 349.6332  0.2120964 9.342013 0.5022506
##           ACF1 Theil's U
```

```
## Training set 0.02264363      NA
## Test set      0.64032865    1.27502
```

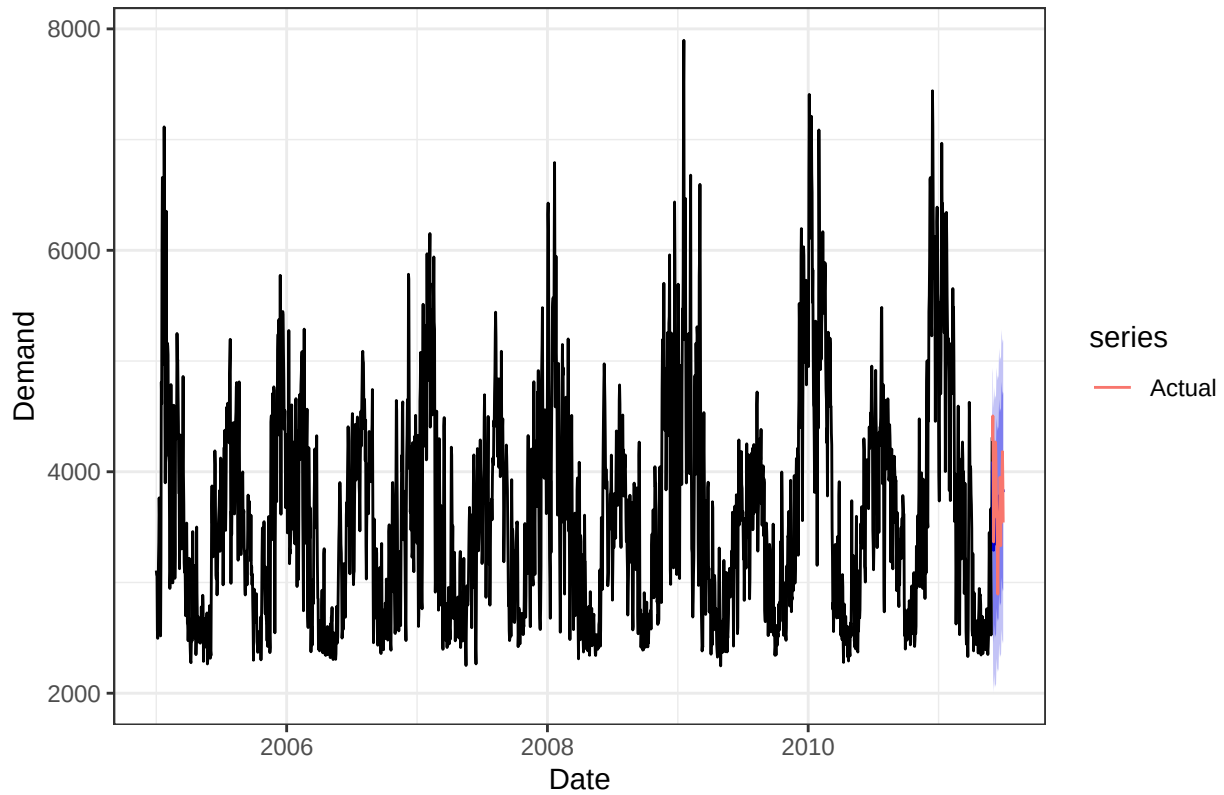
Model 1: TBATS



(2) Model 2: Fourier + ARIMA

```
##           ME      RMSE      MAE      MPE      MAPE      MASE
## Training set  5.726829 428.7071 319.1552 -1.018408 8.722712 0.4584687
## Test set      145.486950 427.0970 348.6946  2.957954 9.138854 0.5009023
##           ACF1 Theil's U
## Training set 0.002445802      NA
## Test set      0.647744049    1.273
```

Model 2: Fourier + ARIMA



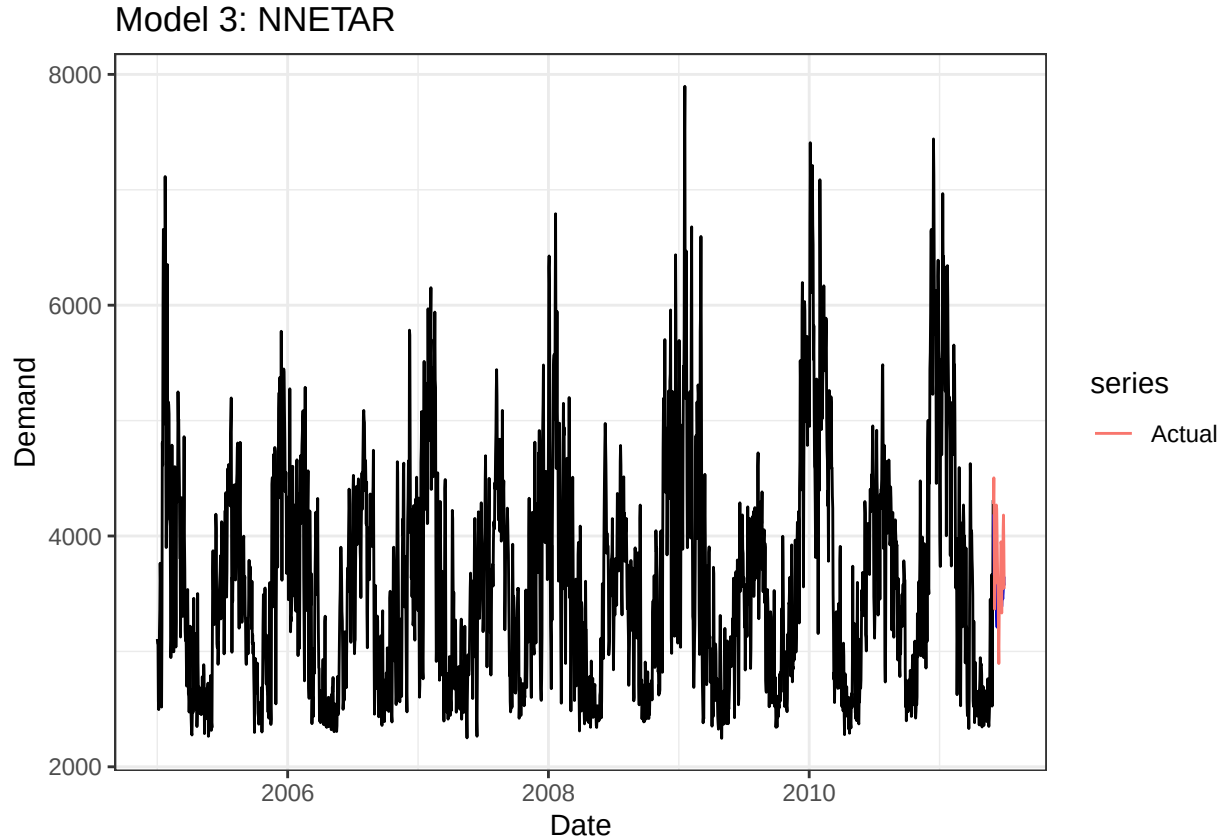
(3) Model 3: NNETAR (Neural Network)

The NNETAR (Neural Network Autoregression) model was implemented to capture potential nonlinear patterns in the electricity load series that may not be fully addressed by traditional linear time series models.

As observed in the results, the NNETAR model is able to follow the general level of the series during the validation period; however, its forecasts appear comparatively smoother and less responsive to sharp spikes in demand. While the model captures the underlying structure to some extent, it tends to underrepresent periods of high volatility and peak load.

	ME	RMSE	MAE	MPE	MAPE	MASE
## Training set	-0.2208004	270.9583	211.7835	-0.8309424	6.038410	0.3042286
## Test set	221.1301396	411.0452	323.3366	5.1016630	8.331915	0.4644753

	ACF1	Theil's U
## Training set	0.001678068	NA
## Test set	0.544305031	1.218031



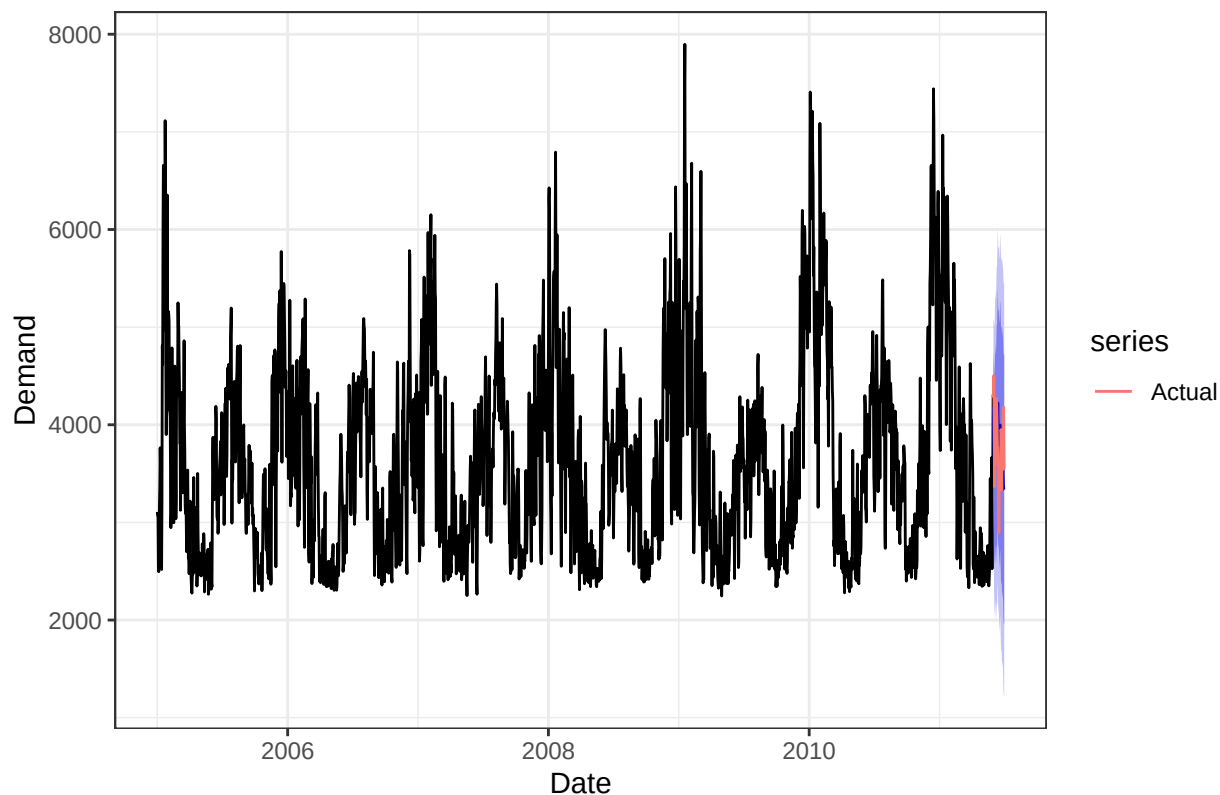
(4) Model 4: STL + ARIMA

The STL + ARIMA approach combines time series decomposition with a parametric forecasting model, where the series is first decomposed into trend and seasonal components using STL, and an ARIMA model is then fitted to the deseasonalized remainder.

Based on the results, the STL + ARIMA model captures the general level of the series reasonably well; however, its forecasts exhibit substantial variability and wide fluctuations, particularly in the prediction interval. In some instances, the model produces unrealistic dips and spikes, indicating instability in capturing short-term dynamics after decomposition.

##		ME	RMSE	MAE	MPE	MAPE	MASE
##	Training set	0.2980849	449.6214	333.8361	-1.056747	9.193842	0.4795579
##	Test set	2.2800409	450.2026	380.1407	-1.023295	10.389922	0.5460748
##		ACF1	Theil's U				
##	Training set	7.302786e-05		NA			
##	Test set	6.972804e-01	1.452389				

Model 4: STL + ARIMA



(5) Model 5: Fourier+ARIMA + TBATS + NNETAR

##	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
## Test set	137.4406	400.3603	325.878	2.757238	8.526779	0.6021127	1.194902

Model Comparison

Through the comparison, Ensemble model has the lowest RMSE. But when submitting to Kaggle, TBATS is the best forecasting model for the competition.

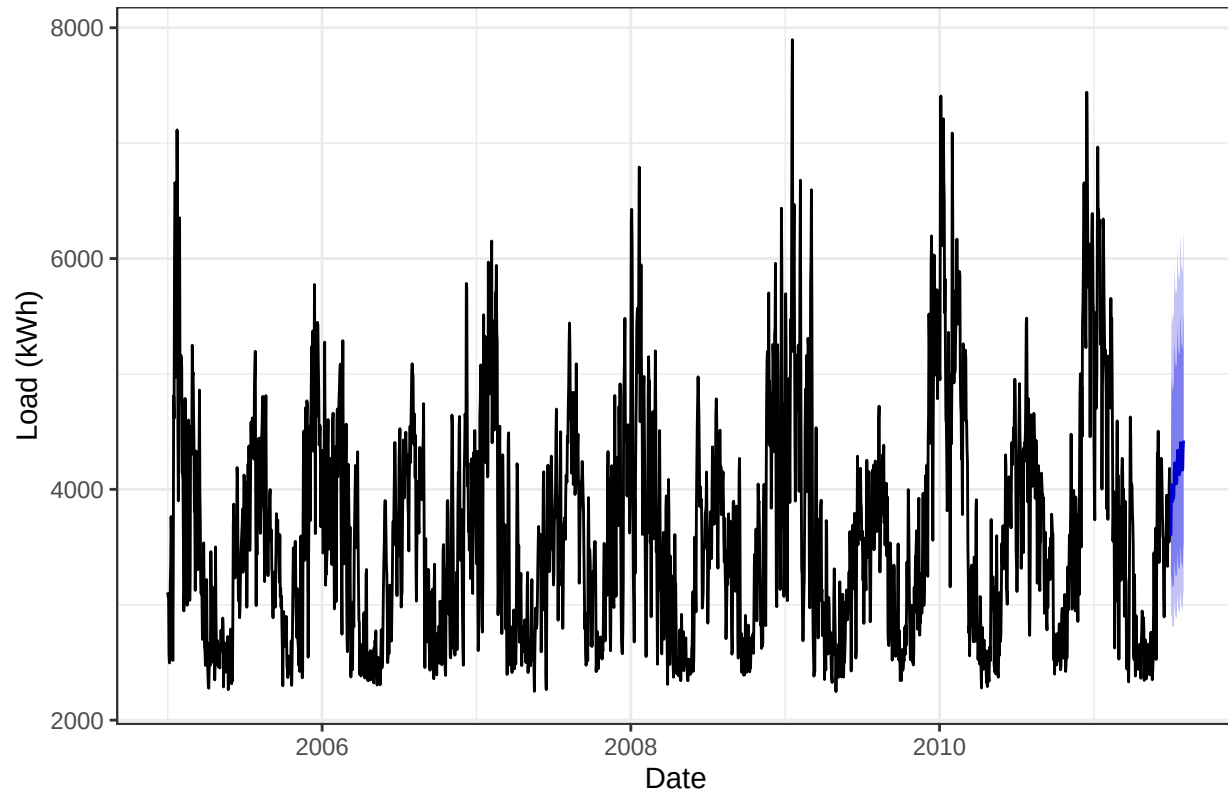
Table 2: Model Comparison: Validation Set (July 2010)

	RMSE	MAPE
Ensemble (TBATS+Fourier+NNETAR)	400.36	8.53
TBATS	417.48	9.34
NNETAR	411.05	8.33
Fourier+ARIMA	427.10	9.14
STL+ARIMA	450.20	10.39

Final

TBATS achieved the best Kaggle public score (0.0998) and was selected as the final model for forecasting daily electricity demand for July 2011.

Final Forecast: July 2011 (TBATS)



References: Other Trials

Beyond the five primary models selected for comparison, a range of additional forecasting approaches were explored to better understand the behavior of the series and assess alternative modeling strategies. These included simple baseline methods such as the arithmetic mean and seasonal naïve (SNAIVE) models, as well as naïve and simple exponential smoothing (SES) applied to a deseasonalized series obtained via STL decomposition. These models provided useful reference points and helped evaluate the impact of explicitly removing seasonal components prior to modeling.

More advanced specifications were also considered, including SARIMA and SARIMAX models, with the latter incorporating exogenous variables such as temperature and humidity to account for external drivers of electricity demand. In addition, hybrid approaches such as STL combined with ETS (STL+ETS) and Fourier-based ARIMA models with weather covariates were tested to better capture multiple seasonalities and nonlinear effects. While some of these models demonstrated competitive performance, they were ultimately not selected as primary models due to either lower predictive accuracy on the validation set or reduced stability compared to the final set of chosen models. Please check Rmd file to check all the models that we tried for forecasting.

AI statement: AI tool was used to help us change the format of date in our data wrangling process.